ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 5

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 60

Marking guideline for the Academic Central Grade 5 Natural Sciences and Technology paper 1 · practice a.

Written by Academic Central for practice, following the CAPS curriculum for Grade 5. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples. Do not take marks off for spelling if the meaning is clear.

QUESTION 1

1 1.1 A ✓ 1.2 B ✓ 1.3 C ✓ 1.4 A ✓ 1.5 A ✓ 1.6 B ✓ 1.7 A ✓ 1.8 C ✓ 1.9 C ✓ 1.10 A ✓ (10)

[10]

QUESTION 2

2 2.1 Sedimentary rock ✓ 2.2 Igneous rock ✓ 2.3 Crust (lithosphere) ✓ 2.4 Hydrosphere ✓ 2.5 Battery ✓ 2.6 Palaeontologist ✓ (6)

[6]

QUESTION 3

3.1 Any TWO: paraffin ✓; gas (LPG) ✓; wood; coal; candle wax; dry dung (2)

3.2 Plants (forests) died millions of years ago ✓. They were buried and pressed under layers of rock and changed into coal over a very long time (heat and pressure) ✓ (2)

3.3 They take millions of years to form, so they will run out (they cannot be replaced) ✓ (1)

3.4 Heat ✓ and light ✓ (2)

3.5 Any TWO: never leave a fire or stove burning alone ✓; keep it away from curtains and things that burn easily ✓; keep paraffin out of reach of children and never in cooldrink bottles; open a window for fresh air; keep water or sand nearby; do not move a stove while it is burning (2)

3.6 Any ONE: it pollutes the air ✓; it causes coughing and lung (breathing) problems such as asthma; it hurts the eyes (1)

[10]

QUESTION 4

4.1.1 A bulb ✓ B cells (battery) ✓ C switch ✓ (3)

4.1.2 Stored chemical energy in the cells → electrical energy ✓ → light energy (and heat energy) ✓ (2)

4.2 Mains, any TWO: kettle ✓; fridge ✓; stove; TV; iron; microwave; heater; washing machine
Cells, any ONE: torch, remote control, clock, toy, radio, calculator ✓ (3)

4.3 Any TWO: never touch plugs or switches with wet hands ✓; do not push objects into plug sockets ✓; do not overload a plug with too many adapters; do not touch broken wires or plugs; never climb pylons or touch fallen power lines; do not make illegal connections (2)

4.4.1 In the wound-up (twisted) elastic band ✓ (1)

4.4.2 Wheels roll, so there is less friction than when you drag (push) the load along the ground ✓ (1)

4.4.3 Any ONE: wheelbarrow ✓; bicycle; car; shopping trolley; wheelchair; skateboard (1)

4.4.4 The cart would not move (or it would only drag and scrape along) ✓ (1)

[14]

QUESTION 5

5.1 (a) hydrosphere ✓ (b) lithosphere ✓ (c) atmosphere ✓ (3)

5.2 Igneous ✓, sedimentary ✓ and metamorphic ✓ (3)

5.3.1 Sedimentary rock ✓ (1)

5.3.2 D ✓. It is at the bottom, and the bottom layers are laid down first ✓ (2)

5.3.3 The fish ✓ (1)

5.3.4 The place was once under the sea (under water) ✓ (1)

5.4 The animal died and was quickly covered (buried) by mud or sand ✓. The soft parts rotted away and only the hard parts (bones, shell, teeth) were left ✓. Over millions of years more layers pressed down, the mud turned into rock and minerals replaced the bones, so they became stone ✓ (3)

5.5 Any ONE: the Cradle of Humankind (Sterkfontein, Maropeng) ✓; the Karoo; the West Coast Fossil Park (Langebaanweg); Golden Gate Highlands National Park (1)

5.6.1 Sand (sandy soil) ✓, clay ✓ and loam ✓ (3)

5.6.2 Loam ✓. It holds enough water but also drains well, and it has humus (food) for plants ✓ (2)

[20]

TOTAL: 60

END OF MARKING GUIDELINE